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CONTINUOUS FUNCTIONS, REVISITED

Prop: In metric spaces (X, d_X) and (Y, d_Y), the ε-δ definition of a continuous function and the topological definition coincide.

Proof:

Standard Examples

Let X, Y and Z be topological spaces.

1. **Constant Functions:** Let $f: X \longrightarrow Y$ be a function s.t. for some $y \in Y$, $f(x) = y$ $\forall x \in X$. Then f is continuous, since the inverse image of any set is either $\emptyset$ or X.

2. **Restricting Domain:** If $f: X \to Y$ is continuous and $A \subseteq X$, then in the subspace topology, the restricted function $f|_A : A \to Y$ is continuous.

   Proof: Let $g = f|_A$. $\forall V \subseteq Y$ open, $g^{-1}(V) = f^{-1}(V) \cap A$, which is open in A.

3. **Restricting / Expanding Range:** Let $f: X \to Y$ be continuous (i) Given a subset Z of Y s.t. $f(X) \subseteq Z$, the function $g: X \to Z$ obtained by restricting the range of f is continuous.
   (ii) If $Y \subseteq Z$, then $h: X \to Z$ obtained by expanding the range of f is continuous.

   Proof: (i) Given $V \subseteq Z$ open in Z, $V = V' \cap Z$ for some open set $V' \subseteq Y$. Then we find $g^{-1}(V) = f^{-1}(V')$, which is an open subset of X.

   (ii) left as an exercise.

4. **Local Formulation:** The map $f: X \to Y$ is continuous if $\exists$ a collection $\{U_\alpha\}_{\alpha \in A}$ of open sets in X s.t. $\bigcup_{\alpha \in A} U_\alpha = X$ and $f|_{U_\alpha}$ is continuous for each α.

   Proof: Assuming such a collection exists, $\forall V \subseteq Y$ open,
   $$f^{-1}(V) = \bigcup_{\alpha \in A} (f^{-1}(V) \cap U_\alpha) = \bigcup_{\alpha \in A} (f|_{U_\alpha})^{-1}(V).$$
   Note: each set $(f|_{U_\alpha})^{-1}(V)$ is open in U_α since $f|_{U_\alpha}$ is cts. Since U_α is open in X, each of these sets is also open in X, and thus their union, $f^{-1}(V)$, is open in X.

5. **Composition:** Let $f: X \to Y$ and $g: Y \to Z$ be continuous. Then $g \circ f: X \to Z$ is continuous.

   Proof: $\forall V \subseteq Z$ open,
   $$(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V)).$$
   $g^{-1}(V) \subseteq Y$ is open, $f^{-1}(g^{-1}(V)) \subseteq X$ is open.

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Homeomorphisms

Def: Let $f: X \longrightarrow Y$ be a bijection. The map f is said to be a "homeomorphism" if f and its inverse $f^{-1}: Y \longrightarrow X$ are both continuous.

Eg:

1. **The simplest linear functions on $\mathbb{R}$:** Consider any function $f: \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = mx + b$ where $m \neq 0$. Then f is a homeomorphism.

2. Consider the function
   $$f: (-1, 1) \longrightarrow \mathbb{R}, \quad x \longmapsto \frac{x}{1 - x^2}$$
   This is a homeomorphism.

   Proof:

3. Let $B_n = \{x \in \mathbb{R}^n : d(x, 0) < 1\}$. Take the function $F: B_n \longrightarrow \mathbb{R}^n$,
   $$x \longmapsto \frac{x}{1 - d(x, 0)}$$
   is a homeomorphism.

   Proof:

Exercise: Show that any two open intervals in $\mathbb{R}$ are homeomorphic.

Non-example: 1. Consider the identity function $f: \mathbb{R}_l \longrightarrow \mathbb{R}_d$: $\{f(x) = x \quad \forall x \in \mathbb{R}\}$. Then f is not continuous: $[a, b)$ is open in $\mathbb{R}_l$, but $f^{-1}([a, b)) = [a, b)$ is not open in $\mathbb{R}_d$.

Exercise: For this f, show that $f^{-1}: \mathbb{R}_d \to \mathbb{R}_l$ is continuous.

2. Let $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ denote the unit circle. Consider $f: [0, 1) \longrightarrow S^1$ given by
   $$f(t) = (\cos(2\pi t), \sin(2\pi t)).$$
   Then f is continuous, bijective, but not a homeomorphism.

   Proof:

Embeddings

Def: Let $f: X \to Y$ be an injective continuous map. Suppose $Z = f(X)$ and consider $f': X \to Z$ obtained by restricting the range of f. Note f' is a bijection.
If f' is a homeomorphism, we say f is a "(topological) embedding" of X in Y.

Eg: 1. Let X, Y be topological spaces, and suppose $y \in Y$. The map $f: X \to X \times Y$ given by $f(x) = (x, y)$ is a topological embedding.

   Proof:

2. Let X be a topological space and A be a subspace of X. The inclusion $i_A: A \longrightarrow X$ is an embedding.

   Proof: Exercise.

Note: Consider $f: [0, 1) \longrightarrow S^1$,
$$x \longmapsto (\cos(2\pi x), \sin(2\pi x))$$
as above. The function $g: [0, 1) \longrightarrow \mathbb{R}^2$ obtained by composing the map of f is not an embedding of $[0, 1)$ in $\mathbb{R}^2$.

Coordinate Functions

Theorem: Let $f: A \longrightarrow X \times Y$ be of the form $f(a) = (f_1(a), f_2(a))$ for some functions $f_1: A \to X$, $f_2: A \to Y$. $\{f_1, f_2\}$ are called the coordinate functions of f. Then f is continuous iff f_1 and f_2 are continuous.

Proof:

Urysohn Lemma Theorem